SCF-Cdc4 complex
FBXW7
SCF-Cdc4 ubiquitin ligase complex
UPS
NDN
AMPK
PROX1
PINK1
PRKN
UPS
nucleus
DNA damage
PARP1
SIRT1
SNCA
S129
ZNF746
PPARGC1A
GSK3B
NAD depletion
T295 S538
T299 T177
MAPK p38
KAT2A
ESRRA
mTORC1
MEF2D
autophagy
CDK5
mitochondrial biogenesis
MEF2C
fatty acid oxidation
SIRT3
COX4I1
COX5A
COX5B
COX6A1
COX6B1
COX6C
COX7A2
COX7B
COX7C
COX8A
CYCS
SDHB
SDHC
SDHD
ATP5F1B
ATP5F1C
complex IV
complex II
complex V
POLRMT
TFAM
TFB2M
TFB1M
mitochondrial transcription
SURF1
VDAC1
TOMM20
mitochondrial transport
RNS
NO
ROS
mutated A53T
CHCHD2
TF CHCHD2
hypoxia
CHCHD4
PARK7
peroxynitrite
CCCP
mitochondrial depolarization
membrane permeability
purine nucleotide
superoxide
H+
SOD1
SIRT5
tiopronin
diazoxide
ATP
ATP-sensitive K+ channel
UCP2
UCP3
mitochondrial dysfunction
Ca2+
rotenone
ubiquinol
NDUFA9
NDUFA10
NDUFA1
NDUFA2
NDUFA3
NDUFA4
NDUFA5
NDUFA6
NDUFA7
NDUFA8
NDUFA11
NDUFA12
NDUFA13
alpha subcomplex
NDUFB1
NDUFB2
NDUFB3
NDUFB4
NDUFB5
NDUFB6
NDUFB7
NDUFB8
NDUFB9
NDUFB10
NDUFB11
beta subcomplex
NDUFS1
NDUFS2
NDUFS3
NDUFS4
NDUFS5
NDUFS6
NDUFS7
NDUFS8
Fe-S protein
NDUFV1
NDUFV2
NDUFV3
MT-ND1
MT-ND2
MT-ND3
MT-ND4
MT-ND4L
MT-ND5
MT-ND6
NDUFAB1
NDUFAF1
NDUFC1
NDUFC2
complex I
UQCRH
UQCRFS1P1
UQCRQ
UQCRC1
UQCR10
UQCR11
UQCRB
MT-CYB
heme b
iron sulfur cluster
complex III
MT-CO1
MT-CO2
MT-CO3
complex IV
ATP5PF
ATP5MG
ATP5PD
ATP5ME
MT-ATP6
MT-ATP8
F(0)
F(1)
complex V
SDHA
Fe-S cluster
FAD
flavoprotein
FADH2
succinic acid
TCA cycle
fumaric acid
OXPHOS assembly factors
AARS2
DARS2
GARS
KARS
RARS2
Mt tRNA synthetases
POLG2
POLG
MTFMT
MTPAP
Mt replication
mtDNA damage
mtDNA replication
SUCLA2
DGUOK
RRM2B
TK2
SUCLG1
Mt-dNTP pool
mtDNA repair
ERCC6
ERCC8
Mt-DNA repair
mitochondrial translation
MRPL3
MRPS16
MRPS22
Mt ribosomal proteins
nucleoid
NDUFAF1
NDUFAF6
NDUFAF2
NDUFAF5
NDUFAF3
ACAD9
NDUFAF4
FOXRED1
CI
COX10
COX14
COX15
FASTKD2
LRPPRC
SCO1
SCO2
TACO1
CIV
SDHAF1
CII
BCS1L
CIII
ATPAF2
TMEM70
CV
ABCB7
BOLA3
NUBPL
NFU1
Fe-S
mitochondrial matrix
mitochondrion
CDK5
MAPK14
MAPK12
PARL
HAX1
HAX1:PARL
HTRA2
S400
S212
LONP1
CLPB
HSPD1
HSPE1
HSPA9
CLPP
proteolysis
mt UPR
TRAP1
chaperone-mediated protein folding
mitochondrial respiratory chain
ribosome
mitochondrion-attached ribosome
&
neuron

AMPK
ribosome
PINK1
PARL
HAX1
HAX1:PARL
MAPK12
Y182
T180
MAPK14
CDK5
ROS
SIRT3
UPS
SCF-Cdc4 complex
FBXW7
SCF-Cdc4 ubiquitin ligase complex
NDN
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mutated A53T
SNCA
S129
SURF1
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VDAC1
NO
TOMM20
autophagy
MAPK p38
CDK5
peroxynitrite
P
N:3
S400
S212
HTRA2
T295
S538
PPARGC1A
T299
T177
mitochondrial transcription
PARP1
SNCA
S129
mitochondrial transport
MEF2D
ZNF746
membrane permeability
mt UPR
PROX1
COX7C
COX6C
COX5B
COX8A
COX4I1
COX7B
COX6A1
COX6B1
COX5A
COX7A2
ATP5F1C
ATP5F1B
SDHD
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SDHC
mTORC1
KAT2A
ESRRA
NAD depletion
PPARGC1A
SIRT1
AARS2
DARS2
GARS
KARS
RARS2
Mt tRNA synthetases
POLG
ERCC6
ERCC8
Mt-DNA repair
TFAM
MRPL3
MRPS16
MRPS22
Mt ribosomal proteins
SUCLA2
DGUOK
RRM2B
TK2
SUCLG1
Mt-dNTP pool
S400
S212
HTRA2
SO2H@C106
PARK7
Ca2+
OXPHOS assembly factors
mitochondrial dysfunction
P
TRAP1
HSPE1
HSPA9
HSPD1
NDUFA9
P
NDUFA10
NDUFA1
NDUFA2
NDUFA3
NDUFA4
NDUFA5
NDUFA6
NDUFA7
NDUFA8
NDUFA11
NDUFA12
NDUFA13
alpha subcomplex
NDUFB1
NDUFB2
NDUFB3
NDUFB4
NDUFB5
NDUFB6
NDUFB7
NDUFB8
NDUFB9
NDUFB10
NDUFB11
beta subcomplex
NDUFS1
NDUFS2
NDUFS3
NDUFS4
NDUFS5
NDUFS6
NDUFS7
NDUFS8
Fe-S protein
NDUFV1
NDUFV2
NDUFV3
MT-ND1
MT-ND2
MT-ND3
MT-ND4
MT-ND4L
MT-ND5
MT-ND6
NDUFAB1
NDUFAF1
NDUFC1
NDUFC2
complex I
CLPP
UQCRH
UQCRFS1P1
UQCRQ
BCS1L
UQCR10
UQCR11
UQCRB
UQCRC1
UQCRC2
UQCRFS1
iron-sulfur cluster
iron-sulfur subunit
MT-CYB
heme b
MT-CYB:heme b
complex III
T295
S538
PPARGC1A
T299
T177
P
P
CYCS
SIRT3
complex IV
complex V
complex II
CHCHD4
SOD1
mtDNA repair
mitochondrial transcription
mitochondrial translation
HSPE1
HSPA9
CLPB
HSPD1
rotenone
myxothiazol
LONP1
ROS
Tiopronin
diazoxide
ATP
chaperone-mediated protein folding
MEF2C
neuron
mitochondrial biogenesis
CHCHD2
superoxide
POLG2
POLG
MTFMT
MTPAP
Mt replication
mtDNA damage
chaperone-mediated protein folding
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NDUFAF6
NDUFAF2
NDUFAF5
NDUFAF3
ACAD9
NDUFAF4
FOXRED1
CI
COX10
COX14
COX15
FASTKD2
LRPPRC
SCO1
SCO2
SURF1
TACO1
CIV
SDHAF1
CII
BCS1L
CIII
ATPAF2
TMEM70
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NUBPL
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Fe-S
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KCNJ11
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MT-CO1
MT-CO2
MT-CO3
COX4I1
COX4I2
COX5A
COX5B
COX6A1
COX6A2
COX7A1
COX7A2L
COX7A2
COX7B
COX7C
COX8A
complex IV
purine nucleotide
mtDNA replication
ATP5IF1
ATP5MC1
ATP5MC2
ATP5MC3
ATP5MF
ATP5PO
ATP5PB
ATP5MG
ATP5PD
ATP5MGL
ATP5ME
MT-ATP6
ATP5PF
MT-ATP8
F(0)
ATP5F1A
ATP5F1B
ATP5F1C
ATP5F1D
ATP5F1E
ATP5F1EP2
F(1)
complex V
SIRT3
GHITM
CHCHD2
reduced
CYCS
CHCHD2:GHITM:CYCS
UCP3
SIRT5
UCP2
SNCA
S129
mutated A53T
SNCA
S129
SLC25A14
SLC25A27
mt UPR
PRKN
Ub
SDHB
Fe-S cluster
Fe-S protein
SDHA
FADH2
flavoprotein
SDHC
SDHD
complex II
H+
ROS
SDHB
Fe-S cluster
Fe-S protein
SDHA
FAD
flavoprotein
SDHC
SDHD
complex II
PARK7
fumaric acid
ubiquinol
succinic acid
H+
CCCP
TCA cycle
mitochondrial depolarization